CSE 219 — Signals and Linear Systems

Lec 6–7 · Laplace Transform: ROC, Properties, Inverse, LTI Systems

Lesson notes · Exam approach · TF practice with solutions · Quick notes (Ashraf Sir's portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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Lesson 15 — Properties of the Laplace Transform — shift, scaling, reversal, convolution, derivative, integration, final value

Lesson 16 — LTI Systems, Differential Equations, Circuits and Feedback — H(s), causality/stability from the ROC, ICs, block diagrams

Quick Notes — Laplace Transform — merged revision notes + key diagrams

NOTE Slide numbers are PDF page numbers of 6 - Laplace.pdf (36 pages, “Lec 6”) and 7 - More Laplace.pdf (28 pages, “Lec 7”); the printed slide number is one less. The slides use the bilateral transform; the unilateral transform (initial conditions at 0^-), final-value theorem, circuits and feedback loops are textbook background (marked) that the EEE papers ask almost every year. Mechanical analogues and state-space questions are not in these slides and are listed in the Outside Notes tab.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note
2023-24 (CSE 219)	Q6(a) y''+7y'+12y = x'+2x, x = u(t), zero IC: solve by LT (12) · Q6(b) LT + ROC of $\|t\|e^{-2\|t\|} - e^{-t}\cos 3t\, u(t) + t^2e^{-4t}u(t)$; sketch ROC (13) · Q6(c) Causal but unstable? stable but non-causal? reason with ROC (10)
2023-24 (EEE 211)	Q4(a) RLC circuit, $v_C(0^-)=1$, $i_L(0^-)=2$, x=u(t): y(t) by LT (18) · Q5(b) Plant 1/(s+2), controller (s+1)/s, unity feedback, x = Au: final value = input? with 1/(s−2)? (15) · Q8(a) Transfer function of the block diagram; poles; stability (17) · Q8(b) y''+5y'+6y = x'+x, $y(0^-)=2$, $y'(0^-)=1$, $x = e^{-4t}u$: ZIR and ZSR by LT (18)
2022-23 (CSE 219)	Q1(b) $x(t) = e^{-at}$ on 0<t<T: is s = −a in the ROC? (apparent pole) (10) · Q2(a) State and prove time- and s-domain differentiation properties of the LT (10) · Q2(b) Inverse LT of $(s+2)/(s^2+7s+12)$, $-4 < \mathrm{Re}\{s\} < -3$ (10) · Q2(c) LT and ROC of $e^{-b\|t-2\|}$ (15) · Q4(a) Relation between the Laplace and the Fourier transform (10)
2022-23 (EEE 211)	Q1(a) Thermostat: choose $H_1 = s^2/(s^2+9)$ or H_2; % incremental change at 1.5 s (18) · Q3(a) Circuit with v = 2 V (t<0), $4e^{-2t}$ (t≥0): $i_R(t)$ by LT (18)
2021-22 (EEE 211)	Q7(a) LT + ROC: $e^{-at}u$ (a>0, a<0), $-e^{-at}u(-t)$ (a<0) (12) · Q7(b) Convolution property: inverse LT of $1/(s-a)^3$ (12) · Q7(c) y'+2y = $e^{-3t}u(t)$, $y(0^-) = 1$ by LT (11) · Q8(a) Control loop with disturbance: H_d = Y/W; zero steady state if H_c has a zero at 0 (18)
2020-21 (EEE 211)	Q3(a) $H_1 = 1/(s^2+bs+(b-3))$, $H_2 = -2$ feedback: overall H, b for stability (15) · Q3(b) Switched circuit (10 V, 4Ω, 0.2F, 3Ω, 6Ω, 0.5H): i(t) by LT (20) · Q4(a) Block diagram $H_1 = 1/(s+2)$, $H_2 = s/(s+3)$, K = 2: H(s) and step response (20)
2019-20 (EEE 211)	Q3 Multipath y = x + a x(t−T): H(s); inverse system as feedback; its h; causal & stable condition (35)
2018-19 (EEE 211)	Q5(a) X(s): two poles, no finite zeros, pole at −1+j, $e^{2t}x$ not abs. integrable, X(0)=5: X(s), ROC (15) · Q5(b) i_s = 2 A (t≤0), $4e^{-2t}$ A (t≥0) into R = 2 Ω ‖ L = 2 H: $i_L(t)$ by LT (20) · Q8(a) y''+3y'+Ky = x'': K for BIBO stability; ZIR for $y(0^+)=0$, $y'(0^+)=1$, K=2 (15)
2017-18 (EEE 211)	Q3(a) s-domain circuit, x = cos t u(t), $v_C(0^-)=1$, $i_L(0^-)=2$: y(t) (17) · Q3(b) $H_1 = 1/(s^2+bs+(b-2))$, $H_2 = -1$: H(s); b for stability (18) · Q4(b) y''+3y'+2y = 2x'+x: transfer function and h(t) (13) · Q4(c) LT of a periodic ±1 square wave (period 2T); ROC (10)
2016-17 (EEE 211)	Q7(a) Bilateral and unilateral LT of $e^{at}u(-t)$, a > 0 (11) · Q7(b) Given X(s): LT of t·x + dx/dt + $\int_0^t x$ (12) · Q7(c) u(t) → δ(t−1) → [δ(t) ‖ $e^{-2t}u(t)$]: Y(s), y(t) (12) · Q8(b) RLC circuit (L=2, R=1, C=2): H(s) and h(t) (12)
2015-16 (EEE 301)	Q7(b) y''+5y'+6y = $2e^{-t}u$, $y'(0^-) = -2$, $y(0^-) = 3$ by LT (13) · Q8(a) Circuit with $v_C(0^-)=1$, $i_L(0^-)=2$, x=u(t): y(t) by LT (15) · Q8(b) Block diagram G_1, G_2, G_3: Y(s) and final value (10)
2014-15 (EEE 301)	Q7(a) y''+4y'+3y = $e^{-3t}u$, $y(0^-)=0$, $y'(0^-)=1$ by LT (15) · Q7(b) Bilateral LT and ROC of $e^{bt}u(-t)$ (7) · Q7(c) Block diagram 1/(s+2), s/(s+3), gain 2: Y(s) (13) · Q8(a) Circuit with initial energy: transformed model, Y(s), y(t) (13)
2013-14 (EEE 301)	Q5(a) Circuit, $v_C(0^-) = ½$, $i_L(0^-) = -1$, x = u(t): Y(s), y(t) (18) · Q6(a) y''+4y'+3y = u(t), $y(0^-)=2$, $y'(0^-)=1$ by LT (13) · Q6(b) Bilateral LT and ROC of cos(at)u(−t) (7) · Q6(c) Block diagram with disturbance: Y(s), lim y(t) (15)
2012-13 (EEE 301)	Q7(a) LT of |z(t)| for the alternating pulse train (10) · Q7(b) State and prove the final-value theorem (5) · Q7(c) Impulse response from input $e^{-2t}u$ and output $(e^{-t}-3e^{-2t})u$ (5) · Q8(b) Circuit, $v_C(0^-)=1$, $i_L(0^-)=2$, x=u: transformed model, Y(s), y(t) (15)
2011-12 (EEE 301)	Q3(a) Causal, poles −2 and 4, x = 1 → y = 0, $h(0^+) = 4$: H(s) (18) · Q3(b) Direct form II diagram → DE; steady-state & transient to 2cos4t u(t) (17) · Q4(a) y''+y'−6y = x'−x (one causal pole), $x = e^{-t}u + e^{t}u(-t-2)$: ROC of Y, y(t) (18) · Q8(a) y''+5y'+6y = x''+5x'+5x: impulse response (17)
2010-11 (EEE 301)	Q3(a) y''+3y'+2y = 0, $y(0^-)=3$, $y'(0^-)=1$: Y(s); H(s) for $x = e^{-3t}u$; final value (12) · Q3(b) Impulse response of $H = (s^2-3s+2)/(s^3+6s^2+11s+6)$; canonical forms (23) · Q4(a) Feedback with r = Au, w = Bu: Y(s); final value (20)

EXAM FOCUS Laplace is the largest topic of Ashraf Sir's part: 32 marks in CSE 219 2022-23 Q1–Q2 and 35 in 2023-24 Q6, and 2–3 questions in every EEE paper. Learn: (1) the ROC rules (right-sided: to the right of the rightmost pole …) and the 'same expression, different ROC, different signal' idea; (2) partial fractions + ROC for the inverse; (3) solving a differential equation with initial conditions at 0^- (ZIR + ZSR); (4) H(s) from block diagrams / circuits, causal-stable tests from the ROC.

Lesson 13 — Definition, ROC and Transforms of Standard Signals

Lec 6 sl. 4–25

For an LTI system with impulse response h, the input e^{st} (any complex $s = \sigma + j\omega$) gives $y = H(s)e^{st}$, $H(s) = \int h(t)e^{-st}dt$ (sl. 5). Generalising to any signal:

$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt = F\{x(t)e^{-\sigma t}\}$ (sl. 6–7: Laplace = Fourier of the weighted signal)

$s = j\omega$ ($\sigma = 0$) $\Rightarrow X(j\omega)$ = Fourier transform — valid only if the ROC contains the $j\omega$-axis

Relation to the Fourier transform (CSE 219 2022-23 Q4(a)): $x(t)e^{-\sigma t}$ has Fourier transform $X(\sigma + j\omega)$; a growing signal such as $e^{2t}u(t)$ has no Fourier transform but has a Laplace transform for $\sigma > 2$.

Basic Pairs and Why the ROC Matters (sl. 9–15)

Signal	X(s)	ROC
$e^{-at}u(t)$	$1/(s + a)$	$\mathrm{Re}\{s\} > -a$
$-e^{-at}u(-t)$	$1/(s + a)$	$\mathrm{Re}\{s\} < -a$
$3e^{-2t}u - 2e^{-t}u$	$(s - 1)/((s+1)(s+2))$	$\mathrm{Re}\{s\} > -1$
$e^{-2t}u + e^{-t}\cos 3t\, u$	$(2s^2 + 5s + 12)/((s+2)(s^2+2s+10))$	$\mathrm{Re}\{s\} > -1$
$\delta(t) - (4/3)e^{-t}u + (1/3)e^{2t}u$	$(s - 1)^2/((s+1)(s-2))$	$\mathrm{Re}\{s\} > 2$
$e^{-b\lvert t\rvert}$, $b > 0$	$-2b/(s^2 - b^2)$	$-b < \mathrm{Re}\{s\} < b$
$\delta(t)$	1	all s
$u(t)$	$1/s$	$\mathrm{Re}\{s\} > 0$

MEMORIZE The algebraic expression alone does not specify the signal: $1/(s + a)$ is $e^{-at}u(t)$ for $\mathrm{Re}\{s\} > -a$ and $-e^{-at}u(-t)$ for $\mathrm{Re}\{s\} < -a$. A transform = expression + ROC. The ROC is the set of σ for which $\int |x|e^{-\sigma t}dt < \infty$.

ROC Properties (sl. 17–25)

#	Property
1	ROC = vertical strips (depends on σ only)
2	ROC of a rational X(s) contains no poles
3	Finite-duration, absolutely integrable x $\Rightarrow$ ROC = entire s-plane (Ex 9.6: $x = e^{-at}$ on (0, T) has $X = (1 - e^{-(s+a)T})/(s + a)$; the 'pole' at $s = -a$ is cancelled — limit = T)
4	Right-sided $\Rightarrow$ ROC is a right half-plane $\mathrm{Re}\{s\} > \sigma_0$
5	Left-sided $\Rightarrow$ ROC is a left half-plane $\mathrm{Re}\{s\} < \sigma_0$
6	Two-sided $\Rightarrow$ a strip $\sigma_R < \mathrm{Re}\{s\} < \sigma_L$ (or empty $\Rightarrow$ no transform)
7	Rational X: ROC bounded by poles or extends to ∞
8	Rational and right-sided $\Rightarrow$ right of the rightmost pole; left-sided $\Rightarrow$ left of the leftmost pole

Example 9.8: $X = 1/((s+1)(s+2))$ has three possible ROCs — $\mathrm{Re} > -1$ (right-sided), $\mathrm{Re} < -2$ (left-sided), $-2 < \mathrm{Re} < -1$ (two-sided). Only the strip contains... none contains the $j\omega$-axis except $\mathrm{Re} > -1$ (which is causal and stable).

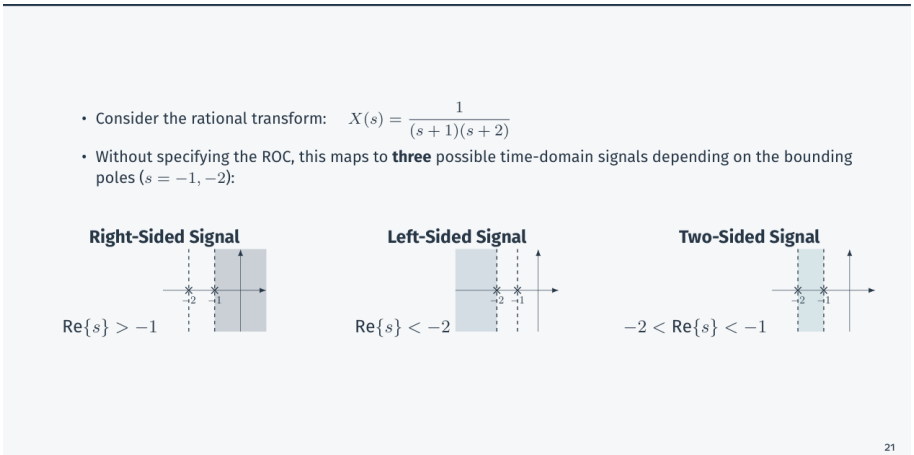


Figure 13.1 — One X(s), three ROCs, three signals (Lec 6, sl. 25)

Exam Approach

1. Split x(t) into right-sided, left-sided and finite parts; transform each with its own ROC.
2. ROC of the sum = intersection of the individual ROCs (empty ⇒ the transform does not exist).
3. Time shift does not change the ROC (e^{-st_0} has no poles); a finite-duration piece has ROC = all s.
4. Sketch the ROC as a shaded strip with the poles (×) marked on its boundaries.

Practice — Lesson 13

(a) Actual — CSE 219 2022-23 Q1(b), Q2(c), Q4(a). (b) Actual — EEE 211 2021-22 Q7(a). (c) Actual — EEE 211 2016-17 Q7(a), EEE 301 2014-15 Q7(b), 2013-14 Q6(b).

(a) (i) x(t) = e^{-at} for 0 < t < T, 0 elsewhere: is s = −a in the ROC of its Laplace transform? (ii) LT and ROC of $e^{-b|t-2|}$ (b > 0). (iii) Relation between the Laplace and Fourier transforms. (10 + 15 + 10)

(b) LT and ROC of (i) $e^{-at}u(t)$, a > 0; (ii) $e^{-at}u(t)$, a < 0; (iii) $-e^{-at}u(-t)$, a < 0. (12)

(c) Bilateral and unilateral LT of $e^{at}u(-t)$ (a > 0); of $e^{bt}u(-t)$; of cos(at)u(−t). (11; 7; 7)

✓ Solution

(a) (i) X(s) = $\int_0^T e^{-(s+a)t}dt = [1 - e^{-(s+a)T}]/(s + a)$. Finite duration and absolutely integrable ⇒ ROC = all s (property 3). At s = −a numerator and denominator both vanish; L'Hôpital: X(−a) = T e^0 = T, finite. So **yes**, s = −a is in the ROC — there is no pole there. (ii) $e^{-b|t-2|} = e^{-b|\tau|}|_{\tau=t-2}$: $e^{-b|t|} = e^{-bt}u(t) + e^{bt}u(-t) \leftrightarrow 1/(s + b) - 1/(s - b) = -2b/(s^2 - b^2)$, −b < Re{s} < b. Time shift by 2 multiplies by e^{-2s}: **X(s) = −2b e^{−2s}/(s² − b²) = 2b e^{−2s}/(b² − s²), −b < Re{s} < b** (shift leaves the ROC unchanged). (iii) X(σ + jω) = $\int[x(t)e^{-\sigma t}]e^{-j\omega t}dt$ = F{x(t)$e^{-\sigma t}$}; setting σ = 0 gives the Fourier transform, valid when the ROC contains the jω-axis; the Laplace transform also exists for signals (growing exponentials) whose Fourier transform does not.

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(b) (i)  1/(s + a), Re{s} > -a (a > 0: ROC contains jω axis ⇒ FT exists)
    (ii) a < 0: e^{-at} = e^{|a|t} grows: 1/(s + a), Re{s} > -a = |a| > 0 (jω-axis not in ROC
⇒ no FT)
    (iii) -e^{-at}u(-t): 1/(s + a), Re{s} < -a = |a| (a < 0 ⇒ ROC Re{s} < |a|, contains the
jω-axis)
(c) bilateral: e^{at}u(-t) ↔ -1/(s - a), Re{s} < a;   unilateral (integral from 0⁻): u(-t) = 0
for t > 0 ⇒ 0
    e^{bt}u(-t) ↔ -1/(s - b), Re{s} < b;   cos(at)u(-t) ↔ -s/(s² + a²), Re{s} < 0
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Lesson 14 — Inverse Laplace Transform

Lec 6 sl. 27–36

Inverse by the Bromwich integral (sl. 27–29): $x(t)e^{-\sigma t} = (1/2\pi)\int X(\sigma + j\omega)e^{j\omega t}d\omega$, hence

$$\mathbf{x(t) = (1/2\pi j)\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}ds, \quad \sigma \text{ anywhere in the ROC}}$$

Evaluated by closing the contour (Cauchy residues) — rarely by hand — or numerically (Riemann sum, sl. 30). In practice: **partial fractions + the ROC to choose each term's signal** (sl. 31–36):

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A/(s + a):  ROC to the right of -a → A e^{-at}u(t);   ROC to the left of -a → -A e^{-at}u(-t)
Ex: X = 1/((s+1)(s+2)) = 1/(s+1) - 1/(s+2)
  Re{s} > -1:        x = (e^{-t} - e^{-2t})u(t)
  Re{s} < -2:        x = (-e^{-t} + e^{-2t})u(-t)
  -2 < Re{s} < -1:   x = -e^{-t}u(-t) - e^{-2t}u(t)
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MEMORIZE Repeated pole: 1/(s + a)^m ↔ $t^{m-1}e^{-at}u(t)/(m-1)!$ (right-sided) — for the left-sided case multiply by $-u(-t)$ instead. If the degree of the numerator ≥ the denominator, divide first (polynomial part ⇒ δ(t), δ′(t)). Complex poles $-\alpha \pm j\beta$: complete the square → $e^{-\alpha t}\cos \beta t$, $e^{-\alpha t}\sin \beta t$.

Exam Approach

1. Factor the denominator, expand in partial fractions (cover-up for simple poles).
2. For each pole decide right- or left-sided from the given ROC (poles to the left of the ROC → right-sided, poles to the right → left-sided).
3. Invert term by term; write the final signal with u(t) and u(−t).
4. Check with the initial value or one numeric value.

Practice — Lesson 14

(a) Actual — CSE 219 2022-23 Q2(b). (b) Actual — EEE 301 2010-11 Q3(b). (c) Practice set 8.1–8.7.

(a) Inverse LT of $X(s) = (s + 2)/(s^2 + 7s + 12)$, $-4 < \text{Re}\{s\} < -3$. (10)

(b) Impulse response of $H(s) = (s^2 - 3s + 2)/(s^3 + 6s^2 + 11s + 6)$ (causal). (6)

(c) Invert $(3s + 7)/((s+1)(s+2))$, Re > −1; $(2s+3)/(s^2+2s+5)$, Re > −1; $-5/((s+1)(s-4))$, $-1 < \text{Re} < 4$; $e^{-s}/(s(s+2))$, Re > 0; $s/(s+2)$, Re > −2.

✓ Solution

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(a) X = (s+2)/((s+3)(s+4)) = -1/(s+3) + 2/(s+4)      [A = (-3+2)/(-3+4) = -1, B = (-4+2)/(-4+3) = 2]
    ROC -4 < Re{s} < -3: pole -3 is to the right of the ROC ⇒ left-sided; pole -4 is to the left ⇒ right-sided:
    -1/(s+3) (Re < -3) ↔ +e^{-3t}u(-t);    2/(s+4) (Re > -4) ↔ 2e^{-4t}u(t)
    x(t) = 2e^{-4t}u(t) + e^{-3t}u(-t)
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(b) H = 3/(s+1) - 12/(s+2) + 10/(s+3)   ⇒  h(t) = (3e^{-t} - 12e^{-2t} + 10e^{-3t})u(t)
[h(0⁺) = 1 ✓: leading coefficient]
(c) (3s+7)/((s+1)(s+2)) = 4/(s+1) - 1/(s+2) ⇒ (4e^{-t} - e^{-2t})u(t)
    (2s+3)/((s+1)²+4) = 2(s+1)/((s+1)²+4) + (1/2)·2/((s+1)²+4) ⇒ e^{-t}[2cos2t + ½sin2t]u(t)
    -5/((s+1)(s-4)) = 1/(s+1) - 1/(s-4); -1 < Re < 4 ⇒ e^{-t}u(t) + e^{4t}u(-t)
    e^{-s}/(s(s+2)) = e^{-s}·½[1/s - 1/(s+2)] ⇒ ½[1 - e^{-2(t-1)}]u(t-1)
    s/(s+2) = 1 - 2/(s+2) ⇒ δ(t) - 2e^{-2t}u(t)
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Lesson 15 — Properties of the Laplace Transform

Lec 7 sl. 4–18

Property	Signal	Transform	ROC
Linearity	$ax_1 + bx_2$	$aX_1 + bX_2$	$\supseteq R_1 \cap R_2$ (pole cancellation can enlarge it: Ex 9.13)
Time shift	$x(t - t_0)$	$e^{-st_0}X(s)$	R
s-shift	$e^{s_0 t}x(t)$	$X(s - s_0)$	$R + \mathrm{Re}\{s_0\}$
Scaling	$x(at)$	$(1/\lvert a\rvert)X(s/a)$	aR
Reversal	$x(-t)$	$X(-s)$	$-R$
Conjugation	$x^*(t)$	$X^*(s)$	R
Convolution	$x_1 * x_2$	X_1X_2	$\supseteq R_1 \cap R_2$
Time differentiation	dx/dt	$sX(s)$	$\supseteq R$
s-differentiation	$-t\,x(t)$	dX/ds	R
Integration	$\int_{-\infty}^{t} x$	$X(s)/s$	$\supseteq R \cap \{\mathrm{Re}\{s\} > 0\}$

MEMORIZE Real x(t): poles and zeros are real or occur in conjugate pairs. $te^{-at}u \leftrightarrow 1/(s + a)^2$, $t^{n-1}e^{-at}u/(n - 1)! \leftrightarrow 1/(s + a)^n$. Example 9.15: $(2s^2 + 5s + 5)/((s+1)^2(s+2)) = 2/(s+1)^2 - 1/(s+1) + 3/(s+2) \Rightarrow x = (2te^{-t} - e^{-t} + 3e^{-2t})u(t)$.

Proofs You May Be Asked For

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Differentiation in time: x = (1/2πj)∫X(s)e^{st}ds ⇒ dx/dt = (1/2πj)∫sX(s)e^{st}ds ⇒ dx/dt ↔
sX(s); ROC ⊇ R
   (u ↔ 1/s, Re > 0;  du/dt = δ ↔ s·(1/s) = 1, all s – the pole cancels and the ROC grows)
Differentiation in s: dX/ds = ∫x(t)(-t)e^{-st}dt ⇒ -t x(t) ↔ dX/ds (ROC unchanged)
Unilateral (initial conditions): L{y'} = sY - y(0⁻);  L{y''} = s²Y - sy(0⁻) - y'(0⁻)
Final-value theorem: if sY(s) has all poles in the open left half-plane, y(∞) = lim_{s→0} sY(s)
   proof: ∫₀^∞ y'e^{-st}dt = sY - y(0⁻) → y(∞) - y(0⁻) as s → 0 ⇒ y(∞) = lim sY(s)
   (does not apply if sY has a pole on the jω-axis other than a simple pole at 0 or in the RHP)
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Exam Approach

1. Identify which properties connect the given signal to a known pair; name each used.
2. Track the ROC through every step (shift moves it, scaling scales it, reversal flips it).
3. Unilateral problems: use the derivative formulas with initial values at 0^-.

Practice — Lesson 15

(a) Actual — CSE 219 2022-23 Q2(a). (b) Actual — CSE 219 2023-24 Q6(b). (c) Actual — EEE 211 2021-22 Q7(b), 2017-18 Q4(c), 2016-17 Q7(b). (d) Actual — EEE 301 2012-13 Q7(a)–(b).

(a) State and prove the time-domain and s-domain differentiation properties of the Laplace transform. (10)

(b) LT and ROC of $x(t) = |t|e^{-2|t|} - e^{-t}\cos(3t)u(t) + t^2e^{-4t}u(t)$; sketch the ROC marking the poles. (13)

(c) Inverse LT of $1/(s - a)^3$ by the convolution property; LT of a periodic ±1 square wave of period 2T; given X(s) of a right-sided signal, find the LT of $y = t\,x + dx/dt + \int_0^t x$.

(d) LT of |z(t)| for the alternating pulse train of height 1, width τ, period 2T (+1 on (0,τ), −1 on (T, T+τ), …); state and prove the final-value theorem.

✓ Solution

(a) Proofs in the box above (time: differentiate the inverse integral under the sign; s: differentiate the defining integral). The ROC of sX(s) contains R; that of dX/ds is exactly R.

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(b) |t|e^{-2|t|} = te^{-2t}u(t) + (-t)e^{2t}u(-t)
    te^{-2t}u(t) ↔ 1/(s+2)²,  Re{s} > -2
    e^{2t}u(-t) ↔ -1/(s-2), Re < 2;  multiply by -t ↔ d/ds: (-t)e^{2t}u(-t) ↔ d/ds[-1/(s-2)] =
1/(s-2)², Re{s} < 2
    ⇒ first term: 1/(s+2)² + 1/(s-2)²,  -2 < Re{s} < 2
    e^{-t}cos3t u(t) ↔ (s+1)/((s+1)²+9), Re{s} > -1   (poles -1 ± 3j)
    t²e^{-4t}u(t) ↔ 2/(s+4)³,  Re{s} > -4
    X(s) = 1/(s+2)² + 1/(s-2)² - (s+1)/((s+1)²+9) + 2/(s+4)³
    ROC: (-2, 2) ∩ (-1, ∞) ∩ (-4, ∞) = -1 < Re{s} < 2
    poles to mark: s = -2 (double, outside the ROC), s = 2 (double), s = -1 ± 3j (on the left
boundary), s = -4 (triple)
    The ROC is the vertical strip between Re = -1 and Re = 2; it contains the jω-axis ⇒
Fourier transform exists.
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(c) 1/(s-a)³: 1/(s-a)² ↔ te^{at}u(t) (right-sided, Re > a); convolution: 1/(s-a)·1/(s-a)²
    e^{at}u * te^{at}u = e^{at}∫₀ᵗ τ dτ·u(t) = (t²/2)e^{at}u(t)   ⇒  x(t) = (t²/2)e^{at}u(t)
    square wave ±1, half-period T: LT of one period = (1 - e^{-sT})²/s ⇒ X = (1 -
e^{-sT})²/(s(1 - e^{-2sT})) = (1 - e^{-sT})/(s(1 + e^{-sT})) = (1/s)tanh(sT/2), Re > 0
    y = t x + dx/dt + ∫₀ᵗx ↔ -X'(s) + sX(s) + X(s)/s  (x causal: x(0⁻) = 0 so L{x'} = sX)
(d) final value: see the box; |z(t)| for the alternating train = rectangular pulses height 1
width τ every T,
    L = (1 - e^{-sτ})/(s(1 - e^{-sT})), Re > 0
```

Lesson 16 — LTI Systems, Differential Equations, Circuits and Feedback

Lec 7 sl. 20–28 · circuits, feedback: textbook

$Y(s) = H(s)X(s)$ with $H = L\{h\}$ the system function (sl. 20). From the differential equation $\Sigma a_k y^{(k)} = \Sigma b_k x^{(k)}$ (zero initial conditions): $H(s) = Y/X = \Sigma b_k s^k / \Sigma a_k s^k$ — zeros from the numerator, poles from the denominator; the equation fixes H(s) but **not the ROC**; causality chooses it (sl. 25–28).

- **Causal** ⇔ ROC is a right half-plane (right of the rightmost pole for rational H).
- **Stable** ⇔ ROC contains the jω-axis. Causal + rational: stable ⇔ all poles in the open left half-plane.
- Ex 9.20: $H = (s - 1)/((s + 1)(s - 2))$: $\mathrm{Re} > 2$ causal-unstable; $-1 < \mathrm{Re} < 2$ stable, non-causal; $\mathrm{Re} < -1$ anti-causal, unstable.
- **Causal but unstable**: $H = 1/(s - 2)$, $\mathrm{Re}\{s\} > 2$ (pole in the RHP). **Stable but non-causal**: $H = -2/(s^2 - 1)$, $-1 < \mathrm{Re} < 1$, $h = e^{-|t|}$ (two-sided).
- Solving: $dy/dt + 3y = x$, $x = e^{-2t}u$: $Y = 1/((s+2)(s+3)) \Rightarrow y = (e^{-2t} - e^{-3t})u$ (sl. 27).

Initial Conditions, ZIR and ZSR (unilateral, textbook)

```
Y(s)·D(s) = N(s)X(s) + I(s)  where I(s) collects the initial-condition terms:
  y″ + a₁y′ + a₀y = ... ⇒ I(s) = (s + a₁)y(0⁻) + y′(0⁻) ;   y′ + ay = ... ⇒ I(s) = y(0⁻)
ZIR = I(s)/D(s) (x = 0);   ZSR = N(s)X(s)/D(s) (zero initial state);   y = ZIR + ZSR
Final value of y: lim sY(s) (poles of sY in the open LHP only)
```

Circuits (textbook)

Replace elements by s-domain models: R → R; L → sL in series with the voltage source $L \cdot i(0^-)$ (or sL in parallel with the current source $i(0^-)/s$); C → 1/(sC) in series with $v(0^-)/s$ (or sC in parallel with the current source $Cv(0^-)$). Then solve by KVL/KCL or voltage/current division. Inputs that are 'on for t < 0' (e.g. 2 V then $4e^{-2t}$ V) set the initial states from the dc steady state at 0^-.

Block Diagrams and Feedback (textbook)

```
cascade: H₁H₂;  parallel: H₁ + H₂;  negative feedback: Y/X = G/(1 + GH)  (H = 1 for unity
feedback)
disturbance W entering after the controller: Y = [G_c G/(1 + G_c G)]R + [G/(1 + G_c G)]W
Stability: poles of the closed loop = roots of 1 + GH = 0.
Steady state: apply the final-value theorem to sY(s) after confirming that the closed-loop
poles are in the LHP.
```

Exam Approach

1. From the DE or diagram get H(s) as a ratio of polynomials; mark poles and zeros.
2. State the ROC from causality (right of rightmost pole) and compare with the jω-axis for stability.
3. Solve with initial conditions: collect terms in Y(s), partial fractions, invert.
4. Circuits: s-domain model with initial-energy sources; use division/KVL; invert.
5. Feedback: reduce with G/(1 + GH); check stability before using the final-value theorem.

Practice — Lesson 16 (A)

(a) Actual — CSE 219 2023-24 Q6(a), Q6(c). (b) Actual — EEE 211 2023-24 Q8(b), Q5(b). (c) Actual — EEE 211 2021-22 Q7(c), 2017-18 Q4(b). (d) Actual — EEE 301 2015-16 Q7(b), 2014-15 Q7(a), 2013-14 Q6(a), 2010-11 Q3(a), 2011-12 Q3–Q4, Q8(a).

(a) $y'' + 7y' + 12y = x' + 2x$, $x = u(t)$, zero initial conditions: solve with the LT. (12) Can a system be (i) causal but unstable? (ii) stable but non-causal? Explain with the ROC and an example. (5 + 5)

(b) $y'' + 5y' + 6y = x' + x$, $y(0^-) = 2$, $y'(0^-) = 1$, $x = e^{-4t}u(t)$: Laplace solution, ZIR and ZSR. (18) Unity-feedback loop, plant $1/(s+2)$, controller $(s+1)/s$, $x = Au$: can the output settle to A? Repeat with controller $1/(s-2)$. (15)

(c) $y' + 2y = e^{-3t}u$, $y(0^-) = 1$; $y'' + 3y' + 2y = 2x' + x$: $H(s)$ and $h(t)$.

(d) $y'' + 5y' + 6y = 2e^{-t}u$, $y'(0^-) = -2$, $y(0^-) = 3$; $y'' + 4y' + 3y = e^{-3t}u$, $y(0^-) = 0$, $y'(0^-) = 1$; $y'' + 4y' + 3y = u$, $y(0^-) = 2$, $y'(0^-) = 1$; $y'' + 3y' + 2y = 0$, $y(0^-) = 3$, $y'(0^-) = 1$ (then H for $x = e^{-3t}u$ and the final value); $y'' + 5y' + 6y = x'' + 5x' + 5x$: $h(t)$.

✓ Solution

```
(a) Y = (s+2)/(s(s+3)(s+4)) = (1/6)/s + (1/3)/(s+3) − (1/2)/(s+4)
    y(t) = [1/6 + (1/3)e^{-3t} − (1/2)e^{-4t}] u(t)      (y(0⁺) = 0 ✓, y(∞) = 1/6 = H(0))
    (i)  causal but unstable: H = 1/(s−2), ROC Re > 2 (right half-plane ⇒ causal; the pole at
+2 ⇒ ROC excludes the jω axis ⇒ unstable): h = e^{2t}u(t)
    (ii) stable but non-causal: H = −2/(s²−1), ROC −1 < Re < 1 (strip contains jω; not a right
half-plane): h = e^{−|t|}
```

```
(b) Y(s²+5s+6) = (s+1)X + (s+5)y(0⁻) + y'(0⁻) = (s+1)/(s+4) + 2(s+5) + 1
    ZIR: (2s+11)/((s+2)(s+3)) = 7/(s+2) − 5/(s+3)        ⇒ (7e^{−2t} − 5e^{−3t})u(t)
    ZSR: (s+1)/((s+2)(s+3)(s+4)) = −½/(s+2) + 2/(s+3) − 3/2/(s+4) ⇒ (−½e^{−2t} + 2e^{−3t} −
1.5e^{−4t})u(t)
    y(t) = (6.5e^{−2t} − 3e^{−3t} − 1.5e^{−4t})u(t)       (y(0⁺) = 2 ✓)
    Feedback: G = (s+1)/(s(s+2)), T = G/(1+G) = (s+1)/(s²+3s+1): poles (−3 ± √5)/2 ⇒ stable;
    y(∞) = lim s·T·A/s = T(0)A = A ⇒ the goal is met (integral action: the controller has a
pole at 0).
    Controller 1/(s−2): T = 1/(s²−3): poles ±√3, one in the RHP ⇒ unstable: the output grows
like e^{√3t};
    the final-value theorem does not apply (the formal limit −A/3 is meaningless).
```

```
(c) y'+2y = e^{−3t}u, y(0⁻) = 1: Y = (1 + 1/(s+3))/(s+2) = 2/(s+2) − 1/(s+3) ⇒ y = (2e^{−2t} −
e^{−3t})u
    H = (2s+1)/((s+1)(s+2)) = −1/(s+1) + 3/(s+2) ⇒ h = (3e^{−2t} − e^{−t})u(t)  (causal,
stable)
```

```
(d) 2e^{−t}: Y = (2/(s+1) + 3s + 13)/((s+2)(s+3)) = 1/(s+1) + 5/(s+2) − 3/(s+3) ⇒ y = (e^{−t}
+ 5e^{−2t} − 3e^{−3t})u
    e^{−3t}: Y = (1/(s+3) + 1)/((s+1)(s+3)) = (3/4)/(s+1) − (3/4)/(s+3) − (1/2)/(s+3)² ⇒ y =
(¾e^{−t} − ¾e^{−3t} − ½te^{−3t})u
    u: Y = (2s² + 9s + 1)/(s(s+1)(s+3)) = (1/3)/s + 3/(s+1) − (4/3)/(s+3) ⇒ y = (1/3 + 3e^{−t}
− (4/3)e^{−3t})u
    free: Y = (3s+10)/((s+1)(s+2)) = 7/(s+1) − 4/(s+2) ⇒ y = (7e^{−t} − 4e^{−2t})u, y(∞) = 0;
          H = 1/((s+1)(s+2)); x = e^{−3t}u: Y_zs = 1/((s+1)(s+2)(s+3)), lim sY_total = 0
    H = (s²+5s+5)/(s²+5s+6) = 1 − 1/((s+2)(s+3)) = 1 − 1/(s+2) + 1/(s+3) ⇒ h = δ(t) + (e^{−3t}
− e^{−2t})u(t)
```

Practice — Lesson 16 (B) — Circuits, Feedback, System Identification

(e) Actual — EEE 211 2023-24 Q4(a), 2022-23 Q3(a), Q1(a), 2020-21 Q3(b), 2018-19 Q5(b), 2017-18 Q3(a); EEE 301 2016-17 Q8(b), 2015-16 Q8(a), 2014-15 Q8(a), 2013-14 Q5(a), 2012-13 Q8(b). (f) Actual — EEE 211 2023-24 Q8(a), 2020-21 Q3(a), Q4(a), 2017-18 Q3(b), 2019-20 Q3, 2018-19 Q5(a); EEE 301 2014-15 Q7(c), 2013-14 Q6(c), 2015-16 Q8(b), 2016-17 EEE 211 Q7(c), 2021-22 Q8(a), 2010-11 Q4(a), 2011-12 Q3(a)–(b), Q4(a), 2012-13 Q7(c).

(e) Series inductor $L = 2$ H feeding the parallel $R = 1\ \Omega$, $C = 2$ F (output $y = v_C$), $v_C(0^-) = 1$ V, $i_L(0^-) = 2$ A, $x = u(t)$: $y(t)$ by LT; $H(s)$ and $h(t)$ of the same circuit.

(f) $H_1 = 1/(s^2 + bs + (b-3))$ with feedback $H_2 = -2$ (or $H_1 = 1/(s^2 + bs + (b-2))$, $H_2 = -1$): overall H and the values of b for stability. $u(t) \to \delta(t-1) \to [\delta(t) \parallel e^{-2t}u(t)] \to y$: $Y(s)$, $y(t)$. Multipath $y = x + a\,x(t-T)$: $H(s)$, inverse system, h_{inv}, condition for causal and stable.

✓ Solution

(e) Elements: L → 2s with series source 2·i(0⁻) = 4 V (opposing), C → 1/(2s) with v_C(0⁻) = 1 V.
KVL/KCL: X/s = 2(sI − 2) + V, I = V/R + C(sV − 1) = (2s+1)V − 2
⇒ V(4s² + 2s + 1) = 1/s + 4s + 4 ⇒ Y(s) = (4s² + 4s + 1)/(s(4s² + 2s + 1)) = 1/s + 2/(4s² + 2s + 1)
4s² + 2s + 1 = 4[(s + ¼)² + 3/16]: y(t) = [1 + (2/√3)e^{−t/4} sin(√3t/4)] u(t)
check: y(0⁺) = 1 = v_C(0⁻) ✓, y(∞) = 1 = x
H(s) = 1/(4s² + 2s + 1) ⇒ h(t) = (1/√3) e^{−t/4} sin(√3t/4) u(t)
(other circuits of this type: same steps — s-domain model, KVL/KCL with initial sources, partial fractions)

(f) $H_1/(1 + H_1H_2)$ with $H_2 = -2$: H = 1/(s² + bs + b − 5); with $H_2 = -1$ and (b − 2): H = 1/(s² + bs + b − 3)
stability (causal, 2nd order): all coefficients > 0 ⇒ b > 5 (first), b > 3 (second) [b > 0 also needed]
u → δ(t−1) → parallel: Y = (1/s)e^{−s}(1 + 1/(s+2)) = e^{−s}(s+3)/(s(s+2))
= e^{−s}[(3/2)/s − (1/2)/(s+2)] ⇒ y = u(t−1)[3/2 − ½e^{−2(t−1)}]
multipath: H = 1 + a e^{−sT}; inverse H_inv = 1/(1 + a e^{−sT}) = feedback system with forward gain 1 and
feedback path a e^{−sT} (delay T, gain a); expanding for |a| < 1: h_inv = Σ_{k≥0}(−a)^k δ(t − kT);
causal and stable (ROC Re{s} > $\sigma_0 \ni j\omega$) iff |a| < 1 (poles of H_inv at e^{−sT} = −1/a lie in the left half-plane).
Block diagrams with H_1, H_2, K or disturbances: reduce with G/(1+GH); disturbance transfer H_d = G/(1 + G_cG),
steady-state error zero for a step disturbance if G_c has a zero at s = 0 ... (H_d(0) = 0 ⇒ G_c(s) ~ s numerator zero)

NOTE Several EEE questions in this tab depend on a figure (block diagram, circuit) that is only partly legible in the scanned papers — they are marked ‘Method’ or ‘Partly’ in the tracker. Method for each: write H(s) from the diagram, check ROC/stability, then partial fractions (and final value for steady-state questions). EEE 301 2012-13 Q7(c): H = Y/X = (s + 2)[1/(s+1) − 3/(s+2)] = (s+2)/(s+1) − 3 = 1/(s+1) − 2, so **h(t) = e^{−t}u(t) − 2δ(t)**.

Quick Notes — Laplace Transform

Revision points for Lessons 13–16.

L13 · ROC

- $X = \int x\, e^{-st}$; LT = FT of $x\, e^{-\sigma t}$; ROC = vertical strip, no poles; finite signal ⇒ all s.
- Right-sided → right of rightmost pole; left-sided → left of leftmost; two-sided → strip.
- $e^{-at}u \leftrightarrow 1/(s+a)$ Re > −a; $-e^{-at}u(-t) \leftrightarrow$ same, Re < −a; $e^{-b|t|} \leftrightarrow -2b/(s^2-b^2)$, |Re| < b.

L14 · Inverse

- Partial fractions; pole left of ROC → right-sided term; pole right of ROC → left-sided.
- $(s+2)/((s+3)(s+4))$, −4 < Re < −3 → $2e^{-4t}u(t) + e^{-3t}u(-t)$.

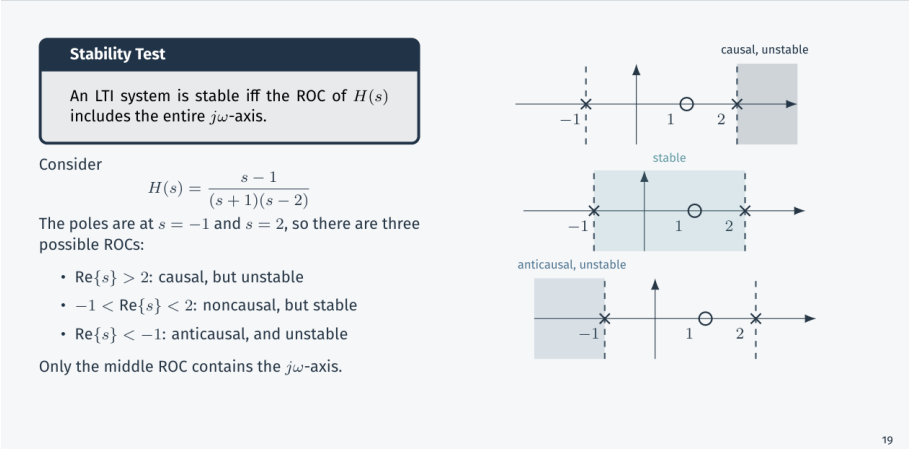
L15 · Properties

- shift $e^{-st_0}X$; s-shift $X(s-s_0)$; scale $X(s/a)/|a|$; d/dt ↔ s; −t x ↔ dX/ds; ∫ ↔ X/s; conv ↔ product.
- Unilateral: sY − $y(0^-)$; final value lim sY.

L16 · Systems

- H = N/D; causal ⇔ right half-plane ROC; stable ⇔ ROC ∋ jω.
- y = ZIR + ZSR; feedback G/(1+GH). 6(a) CSE 219: $y = [1/6 + e^{-3t}/3 - e^{-4t}/2]u$.

Key Diagrams to Reproduce



Stability from the ROC: three ROCs of (s−1)/((s+1)(s−2)) (Lec 7, sl. 22)